

ALLEN AAF, AK, USA

WMO: 702670

Lat: 63.994N

Lon: 145.721W

Elev: 389

StdP: 96.74

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26415

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.3	-34.7	-36.3	0.1	-32.9	-34.4	0.2	-30.9	18.7	-16.2	17.5	-15.2	0.9	170	0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	25.8	14.9	24.0	13.9	22.2	13.3	16.1	23.4	15.1	21.8	14.2	20.2	3.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	9.9	17.3	12.3	9.3	16.5	11.4	8.8	16.0	46.0	23.8	43.4	21.8	40.9	20.2	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.3	(b) 13.6	(c) 12.0	(e) -41.4	(f) 28.6	(g) 3.0	(h) 1.9	(i) -43.6	(j) 29.9	(k) -45.4	(l) 31.0	(m) -47.1	(n) 32.0	(o) -49.2	(p) 33.4		
(4) 15.3	(5) 13.6	(6) 12.0	(7) -41.4	(8) 28.6	(9) 3.0	(10) 1.9	(11) -43.6	(12) 29.9	(13) -45.4	(14) 31.0	(15) -47.1	(16) 32.0	(17) -49.2	(18) 33.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.9	-17.8	-13.0	-9.6	1.0	9.1	14.6	15.7	13.0	7.3	-2.4	-13.1	-15.9	
	DBStd	13.78	10.02	8.71	7.84	5.72	4.39	3.26	2.89	3.57	4.50	6.17	7.85	8.48	
	HDD10.0	4450	861	645	607	271	69	3	2	11	102	386	692	802	
	HDD18.3	7022	1119	878	866	520	286	119	88	168	332	643	942	1060	
	CDD10.0	487	0	0	0	1	42	140	179	104	19	1	0	0	
	CDD18.3	19	0	0	0	0	1	7	8	4	0	0	0	0	
	CDH23.3	216	0	0	0	0	11	73	81	51	0	0	0	0	
	CDH26.7	27	0	0	0	0	1	11	9	7	0	0	0	0	
Wind	WSAvg	4.3	5.5	5.3	4.4	4.1	4.0	3.4	3.2	3.1	3.8	4.2	4.9	5.6	
Precipitation	PrecAvg	292	8	6	5	6	23	59	70	52	26	16	12	8	
	PrecMax	450	27	18	28	17	78	138	152	122	49	53	35	26	
	PrecMin	195	0	0	0	0	2	8	18	14	4	0	0	0	
	PrecStd	74	7	5	6	5	17	33	32	24	12	12	10	7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	6.4	9.3	17.6	24.6	28.4	28.0	27.7	19.5	14.5	5.3	6.0	
		MCWB	2.8	2.0	3.8	8.1	11.8	15.2	16.1	15.8	11.1	8.5	1.9	2.2	
	2%	DB	2.2	2.9	6.1	13.8	21.9	25.4	25.6	24.7	17.6	10.3	1.0	0.9	
		MCWB	-0.7	-0.7	1.5	6.0	10.6	14.1	15.1	15.0	10.0	5.3	-0.8	-1.6	
	5%	DB	-1.8	0.2	4.0	11.2	19.5	23.4	23.8	21.9	16.0	8.1	-1.1	-2.6	
		MCWB	-3.5	-2.2	0.0	4.4	9.7	13.4	14.5	13.8	9.3	3.8	-2.5	-4.2	
	10%	DB	-4.6	-2.4	1.6	9.4	17.2	21.4	22.1	19.7	14.5	5.6	-3.1	-4.8	
		MCWB	-5.8	-4.2	-1.7	3.3	8.7	12.5	13.9	12.7	8.6	2.5	-4.4	-6.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	2.3	3.9	8.6	12.8	16.9	17.3	17.1	11.7	8.6	1.8	2.4
			MCDB	6.5	6.4	9.2	16.6	21.8	26.1	25.2	25.9	17.6	14.2	5.4	5.7
2%		WB	-0.5	-0.3	1.7	6.3	11.5	15.3	16.1	15.8	10.7	5.7	-0.5	-1.3	
		MCDB	2.4	2.8	5.6	13.5	20.1	22.9	22.8	23.1	16.0	9.7	0.9	0.8	
5%		WB	-3.2	-2.0	0.2	4.7	10.3	14.2	15.3	14.5	10.0	4.1	-2.4	-4.0	
		MCDB	-1.5	0.4	3.8	10.6	18.5	21.2	21.7	20.5	14.9	7.6	-1.1	-2.6	
10%		WB	-5.6	-4.1	-1.6	3.5	9.1	13.2	14.6	13.3	9.1	2.6	-4.4	-6.0	
		MCDB	-4.3	-2.3	1.6	8.9	16.4	19.9	20.6	18.2	13.7	5.4	-3.2	-4.7	
Mean Daily Temperature Range		5% DB	MDBR	6.9	8.0	10.7	10.3	10.2	9.6	8.9	9.1	8.6	6.7	6.5	6.7
			MCDBR	8.8	8.5	10.2	12.7	13.7	12.9	12.7	12.8	11.0	8.2	7.4	9.3
	MCWBR		6.9	6.2	7.0	7.1	6.2	5.3	5.3	5.7	5.7	4.9	5.8	7.4	
	5% WB	MCDBR	8.8	8.3	9.8	12.1	12.4	11.2	10.6	11.4	9.8	7.6	7.2	9.5	
		MCWBR	7.0	6.2	6.7	7.0	5.7	4.9	4.8	5.4	5.4	4.8	5.6	7.7	
Clear-Sky Solar Irradiance	taub	0.203	0.266	0.288	0.336	0.359	0.378	0.381	0.390	0.306	0.274	0.199	0.198		
	taud	1.938	2.122	2.180	2.240	2.334	2.316	2.341	2.297	2.521	2.387	1.970	1.910		
	Ebn at Noon	367	644	798	836	850	837	822	772	797	658	362	141		
	Edh at Noon	28	62	89	107	107	112	107	102	66	48	26	12		
All-Sky Solar Radiation	RadAvg	0.19	0.88	2.64	4.59	5.34	5.40	4.71	3.72	2.32	1.02	0.28	0.07		
	RadStd	0.02	0.08	0.18	0.23	0.34	0.42	0.32	0.31	0.24	0.12	0.03	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+72	N/A
(47) Regional (5 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	-326	-354	N/A

Nomenclature: See separate page